



UNIVERSITY
of HULL

Computer Science

BSc / MEng Robotics and Artificial Intelligence

October 2026

Robotics and Artificial Intelligence

Shaded modules are offered as options on your programme, all other modules are compulsory.

Level 4 (Certificate Stage)

Trimester 1	Trimester 2
Programming Portfolio (40cr. core)	
Algorithms and Data Structures	Databases and the Cloud
Professional Development	Computational Thinking

Common first year allows for easy transfer between the available programmes.

Level 5 (Diploma Stage)

Trimester 1	Trimester 2
Robotics Foundations (40cr. core)	
Artificial Intelligence	Data Analysis and Visualisation
Electronics and Microcontrollers	Connected Devices and IoT

Post-Diploma Stage (optional)

Trimester 1	Trimester 2
Year in Industry/Year Abroad	

Level 6 (Honours Stage)

Trimester 1	Trimester 2
Honours Stage Project (40 credits)	
Communicating and Teaching Computing (40 credits)	
Computer Vision	Autonomous Robots
Machine Learning	Embedded Systems Applications

Level 7 (Masters Stage) (optional)

Trimester 1	Trimester 2
Commercial Development Practice (40 credits)	
Edge AI	Cutting Edge Concepts
Dependable Reactive Real-Time Systems	Responsible, Safe and Explainable AI

Please note that the content in the following pages is subject to change, as our curriculum evolves to match the rapidly changing Computer Science landscape.

Level 4

Programming Portfolio

Core tools: C#

Whether you've never written a line of code or you've been tinkering for years, this module meets you where you are. You'll learn to think like a programmer — breaking problems down, writing clean logic, and building working software in C#. By the end of the year you'll have moved from your very first program to designing with classes, objects and inheritance.

You'll explore:

- Core programming building blocks — variables, loops, functions and data types
- Searching, sorting and working with arrays, stacks and queues
- Object-oriented programming — classes, inheritance and encapsulation
- Reading, writing and managing code like a professional, using source control

Algorithms and Data Structures

Every great piece of software starts with a great idea for how to solve the problem — that's what this module teaches you to find. You'll step away from the screen and learn to think in code before you write it, exploring the algorithms and data structures that separate software that merely works from software that works brilliantly.

You'll explore:

- Pseudocode and state machines for planning before you program
- Core data structures, and how to choose the right one for the job
- Sorting and searching algorithms, and how to measure their efficiency
- Design thinking that carries into every other module on your degree

Databases and the Cloud

Core tools: C#, Microsoft SQL Server

Modern software rarely stores its data on just one machine anymore — this module shows you how databases and the cloud work together to make that possible. You'll learn how cloud computing and virtualisation actually work, then design and build real databases, from relational systems through to NoSQL, before bringing the two together to build a working application backed by a database hosted in the cloud.

You'll explore:

- Designing and implementing relational and NoSQL databases
- Virtualisation, containers, and the cloud service models behind them — IaaS, PaaS and SaaS
- Connecting a real application to a cloud-hosted database

- Cloud security, reliability and disaster recover

Computational Thinking

Core tools: programming platforms and mathematical software

Computing is bigger than any one programming language — this module is about how to think as a computer scientist. Drawing on a genuinely global history of innovation, you'll build the mathematical and logical foundations that everything else on your degree rests on.

You'll explore:

- Abstraction and computational methods
- The maths behind computing — algebra, number systems and statistics
- Logic, Boolean algebra and set theory
- Vectors and matrices, and their uses in graphics, AI and data
- Practical aspects of how computers handle different number types and the way that mathematical libraries and tools can help to do the actual calculations

Professional Development

A computer science degree is about more than code — this module helps you build the career to go with it. You'll explore the ethical, legal, social and sustainability challenges facing the sector, and the professional skills that set graduates apart. A hands-on team project puts those skills into practice straight away, while you start building a portfolio that showcases your work.

You'll explore:

- Sector challenges — ethics, diversity, sustainability and cybersecurity
- Communication, research and self-presentation skills
- Real teamwork on a programme-specific group project
- Building your own professional portfolio

Level 5

Robotics Foundations

Before you can build a robot that moves intelligently, you need to understand how it moves at all — that's where this module starts. You'll get grounded in the mechanics, electronics and software that bring robots to life, simulating and building real robotic systems from day one.

You'll explore:

- Robotics architecture and the sense-plan-act paradigm
- Mechanics — forces, moments and stress analysis
- Motors, actuators and drive systems
- Simulating and programming robots in ROS 2 and Gazebo, with hands-on hardware

Artificial Intelligence

Learn to build the kind of intelligent systems that are reshaping every industry and understand the ethics behind them. You'll get a practical grounding in how AI actually works, from the history of the field to the algorithms powering it today, building your own solutions to real problem-solving challenges along the way.

You'll explore:

- The history and ethics and trustworthiness of artificial intelligence
- Pathfinding and problem-solving search algorithms
- Supervised and unsupervised learning, and neural networks
- Reasoning under uncertainty with Bayesian methods, and an introduction to reinforcement learning
- A brief introduction of fairness, explainability and AI Safety

Data Analysis and Visualisation

Data is only powerful once someone can make sense of it — this module teaches you to be that person. You'll work with real-world data from cleaning and wrangling it through to building and evaluating models, finishing with the skills to turn numbers into a story that's clear, compelling and honest.

You'll explore:

- Data cleaning, exploration and dimensionality reduction with pandas, scikit-learn and NumPy
- Classification, regression and clustering algorithms
- Rigorous experimentation, evaluation and hyperparameter tuning
- Spotting and mitigating bias in data and models
- Visualising data for maximum impact

Electronics and Microcontrollers

Software needs hardware to run on — this module teaches you to build that hardware yourself. From first principles of voltage and current through to programming your own microcontroller, you'll get genuinely hands-on, building and testing real circuits on a breadboard.

You'll explore:

- Circuits, resistance, capacitors and semiconductors
- Microcontroller programming with Arduino
- Sensors and communication protocols like I2C and SPI
- Practical, hands-on electronic construction

Connected Devices and IoT

From smart homes to industrial sensors, the Internet of Things is everywhere — learn to build the systems that connect it all. You'll explore the communication technologies that let devices talk to each other and to the cloud, and learn to choose the right one for a given real-world application.

You'll explore:

- Wired and wireless interfacing technologies, from Bluetooth to Ethernet
- Publish/subscribe architectures and real-time communication
- Edge and cloud computing for connected devices
- Security and business considerations for IoT systems

Year in Industry/Year Abroad

Take everything you've learned so far and put it to work — in a real workplace or a different country altogether. Spend a year gaining genuine professional experience on an industry placement, or broaden your horizons studying at an overseas institution. Either route builds transferable, employability-boosting skills in a credit-bearing year that counts towards your degree.

You'll explore:

- A year of real workplace experience, tailored to your placement
- Or a year studying abroad, guided by department staff to match your goals

Level 6

Honours Stage Project

This is your chance to lead a project entirely your own — from first idea to final report. You'll choose your own topic and see it through from planning to delivery, supported by videos, live sessions and workshops covering everything you need to succeed.

You'll explore:

- Project planning and literature review
- Experimental design, testing and evaluation
- Research ethics and data protection
- Professional report writing and presentation

Communicating and Teaching Computing

Think you understand computer science? Try explaining it to someone else — that's the real test. Spend real time in schools or colleges as a computing ambassador, supported by workshops on teaching practice and the National Curriculum, and discover how explaining your subject deepens your own understanding of it.

You'll explore:

- Classroom management and lesson planning

- How computer science is taught in schools today
- Reflective, experience-based learning from your own placement
- Effective communication skills – a critical life skill whatever career you choose

Autonomous Robots

Give a robot the ability to think for itself — this module teaches you how. You'll dig into the real challenges of controlling robots autonomously: noisy sensor data, layered software architectures, and the gap between simulation and the real world. The practical element is central — you'll build a working hierarchy of systems that lets a robot sense, decide and act for itself.

You'll explore:

- Control architectures for autonomous behaviour
- Sensor fusion and distributed processing
- Building and testing real robot behaviours
- Bridging simulation and real-world robotics

Computer Vision

Teach a computer to see — and to actually understand what it's looking at. From basic image processing through to recognising objects and interpreting entire scenes, you'll build the skills behind everything from facial recognition to self-driving cars.

You'll explore:

- Image processing and feature extraction
- Segmentation, shape analysis and motion tracking
- Object and activity recognition
- The real-world applications and ethics of computer vision

Embedded Systems Applications

Take a real physical system and give it a brain — this module is as hands-on as computer science gets. You'll design, build and deploy control logic for a genuine embedded system, working at the point where hardware and software meet.

You'll explore:

- Embedded processor architecture and assembly programming
- Hardware-software integration and peripheral interfacing
- Designing control systems, from bang-bang to PID controllers
- Building a complete real-time embedded control system

Machine Learning

Move beyond using machine learning models to really understanding how they work. Building on your foundational AI knowledge, you'll dig into the algorithms behind modern machine

learning, from classic techniques through to the convolutional networks and generative AI shaping the field today.

You'll explore:

- Gradient descent, regression and regularisation
- Decision trees, random forests and clustering
- Convolutional neural networks
- An introduction to generative AI

Level 7

Commercial Development Practice

Build real software for a real client with industry-standard project management techniques and commercial tools. In Commercial Development Practice, the School's own commercial software development unit, you'll work on genuine client projects that build the business and technical skills employers tell us are actively looking for.

You'll explore:

- Requirements analysis, defining specifications and system design
- Team-based implementation and quality control
- Client liaison, negotiation and project management
- Professional, reflective practice

Cutting Edge Concepts

The textbooks can't keep up with computer science — so this module looks at what's happening right now. You'll work with staff to explore genuinely current research across AI, games, security and beyond, learning to critically evaluate academic sources and separate rigorous research from the rest.

You'll explore:

- Identifying high-quality academic sources
- Critically reviewing current research
- Exploring topics beyond the core curriculum
- Skills that carry straight into postgraduate study or research careers

Dependable Reactive Real-Time Systems

Core tools: HiP-HOPS and SafeML for dependability analysis and understanding when machine-learning may no longer be trusted.

In avionics, autonomous vehicles and robotics, systems often have only one chance to get things right. This module looks at how complex software-intensive systems react and respond in real time, tolerate faults and remain safe when things go wrong.

You'll explore:

- Specification and modelling of reactive real-time systems
- Concurrency, communication and resource management
- Hazard identification and risk analysis
- Model-based dependability assessment.
- Fault tolerance, resilience and recovery

Edge AI

AI doesn't just live in the cloud — learn to run it on the device, in real time, wherever it's needed. You'll design intelligent systems that work under real-world constraints of power, memory and connectivity, building and deploying AI models directly onto edge hardware and connecting them through IoT.

You'll explore:

- Real-time data processing on edge-capable hardware
- Lightweight AI models for devices like Raspberry Pi and Jetson
- IoT integration and sensor networks
- A full end-to-end edge AI project

Responsible, Safe and Explainable AI

The most powerful AI in the world is worthless if nobody can trust it — this module teaches you to build AI people can rely on. You'll explore the cutting edge of responsible, explainable AI, examining how fairness, safety, explainability and transparency can be built into systems from the ground up, not bolted on afterwards.

You'll explore:

- Explainable AI (XAI) and interpretable machine learning
- Detecting and mitigating bias in AI systems
- AI safety, robustness and adversarial attacks
- Responsible multi-agent and autonomous AI systems
- Federated and privacy-preserving learning
- Building AI that is fair, transparent, accountable and trustworthy